



BARKER COLLEGE

TRIAL HIGHER SCHOOL CERTIFICATE 1995

MATHEMATICS

AM TUESDAY 22 AUGUST

3/4 UNIT

TIME: 2 HOURS

[Plus 5 minutes' Reading Time]

DSK

BJR

JCS

BHC

BTP

JM*

125 copies

DIRECTIONS TO CANDIDATES:

1. Write your Candidate Number on **EVERY** page.
2. Start each question on a **NEW** page.
3. **ALL** questions are of equal value.
4. Show **ALL** necessary working. Marks may be deducted for careless or badly arranged work.
5. Standard integrals are provided at the end of the paper.
6. Board-approved calculators may be used.

* * * *

QUESTION 1.

(a) Solve for x : $\frac{1}{x-1} \leq 3$ [3m]

(b) Find the exact value of $\int_1^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$ [2m]

(c) Find the derivative of $y = \tan^{-1} 2x$ [2m]

(d) Evaluate $\int_0^{\frac{\pi}{6}} \sin^2 2x dx$ [3m]

(e) (i) Sketch the graph of $y = \sin 2x$ for $0 \leq x \leq 2\pi$ [1m]

(ii) Sketch the line $y = \frac{1}{2}$. Without solving the equation $\sin 2x = \frac{1}{2}$, how many solutions are there for the domain $0 \leq x \leq 2\pi$? [1m]

QUESTION 2. [START A NEW PAGE]

(a) $\int_0^1 \frac{x}{1+x} dx$ (using the substitution $u = 1+x$)

(b) Given the function $y = 3 \cos^{-1}\left(\frac{x}{2}\right)$:

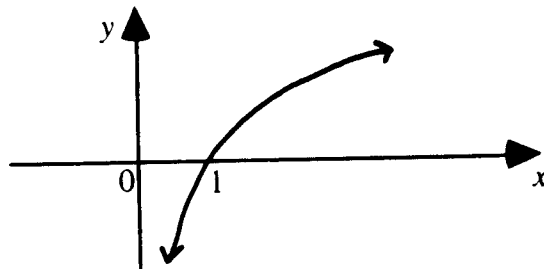
(i) Write down the domain and range.

(ii) Sketch this function.

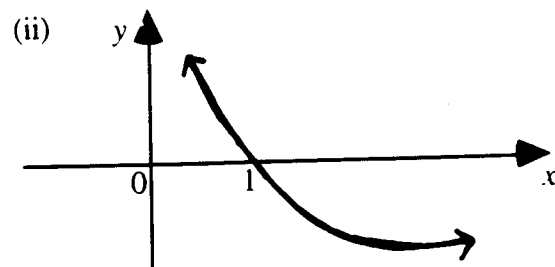
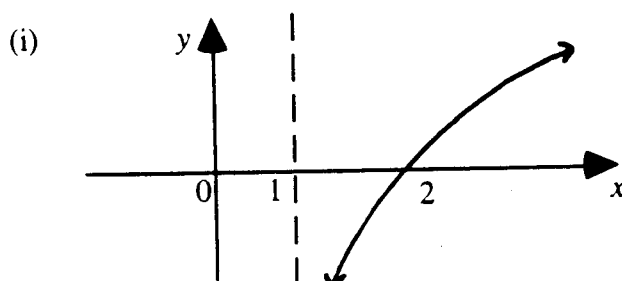
(c) (i) Express $\sin A$ and $\cos A$ in terms of " t " where $t = \tan \frac{A}{2}$

(ii) Hence or otherwise prove that: $\frac{1 + \cos 2A}{\sin 2A} = \cot A$

(d) Given that the following is a sketch of $y = \ln x$



Write down a possible equation for each of the following.



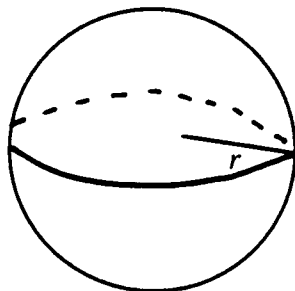
QUESTION 3. [START A NEW PAGE]

- (a) Given the function $y = \frac{2x+1}{x-1}$
- (i) find the domain of this function
 - (ii) what happens to y when $x \rightarrow \infty$
 - (iii) find any vertical and horizontal asymptotes
 - (iv) hence sketch (without calculus) a neat graph of the function. [4m]
- (b) Find the term independent of x in the expansion $\left(2x^3 - \frac{1}{x}\right)^{12}$ [4m]
- (c) Solve the trigonometric equation $2\sin^2 \theta + \sin^2 2\theta = 2$ for $0 \leq \theta \leq 2\pi$ [4m]

QUESTION 4. [START A NEW PAGE]

- (a) Find the acute angle between $2x - y + 5 = 0$ and $y = -3x + 7$ [3m]
- (b) The velocity v m/s of a point moving along the x -axis is given by $v^2 = 16x - 4x^2 + 20$
- (i) Prove that the motion is simple harmonic
 - (ii) Find the centre of motion
 - (iii) Find the length of the path. [4m]

(c)



A spherical balloon is being inflated and its volume increases at a constant rate of 50 mm^3 per second.

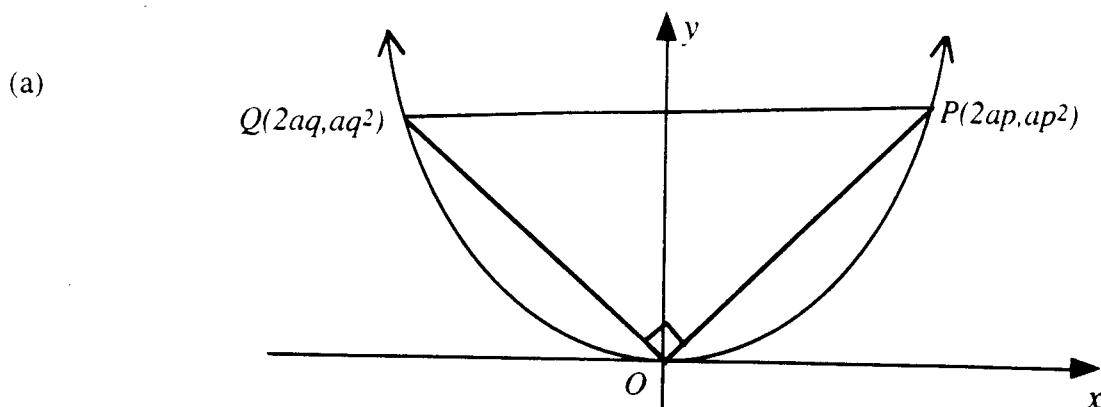
At what rate is its surface area increasing when the radius is 20 mm?

$$V = \frac{4}{3}\pi r^3$$

$$A = 4\pi r^2$$

[5m]

QUESTION 5. [START A NEW PAGE]



PQ is a variable chord of the parabola $x^2 = 4ay$.

It subtends a right angle at the vertex O

If p and q are the parameters corresponding to the points P , Q respectively :

(i) Show that the equation of the tangent to $x^2 = 4ay$ at P is $y - px + ap^2 = 0$

(ii) Hence write down the equation of the tangent at Q , and then find R , the point of intersection of the two tangents drawn from P and Q .

(iii) Find the gradients of PO and QO and hence prove $pq = -4$

(iv) Show that the locus of this point of intersection is $y = -4a$

(b) Use mathematical induction to prove that for all positive integers:

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{1}{6} n(n+1)(2n+1)$$

QUESTION 6. [START A NEW PAGE]

(a) A touring cricket side of 15 players contains 5 regular bowlers.

- (i) How many different elevens can be picked which contain exactly 3 of the 5 regular bowlers?
- (ii) What is the probability that if an eleven is picked at random it will only contain 1 regular bowler?
- (iii) What is the probability that if an eleven is picked at random it will contain at least 3 of the regular bowlers? [5m]

(b) (i) Write down the expansion for $(1+x)^n$

(ii) Using this expansion, show that:

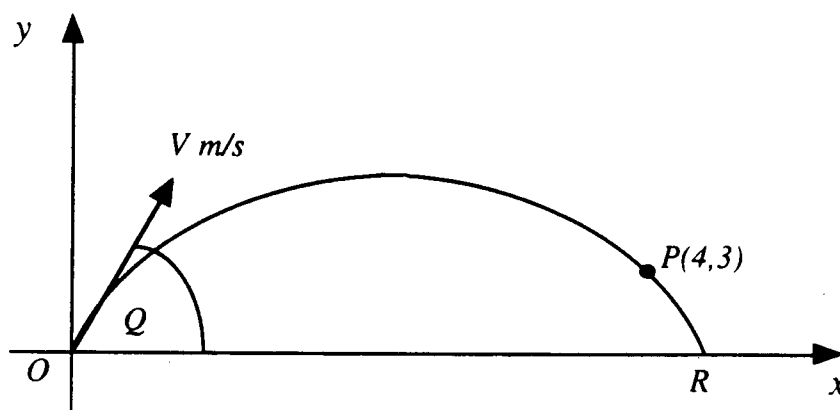
$$\binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \dots = 2^{n-1}$$

(iii) From the identity $(1+x)^n(1+x)^n = (1+x)^{2n}$ compare the coefficient of x^{n+1} on both sides and hence prove that:

$$\binom{n}{0}\binom{n}{1} + \binom{n}{1}\binom{n}{2} + \binom{n}{2}\binom{n}{3} + \dots + \binom{n}{n-1}\binom{n}{n} = \frac{(2n)!}{(n-1)!(n+1)!} \quad [7m]$$

QUESTION 7. [START A NEW PAGE]

- (a) How many 'words' can be formed from AUSTRALIA? (taken all at a time) [2]
- (b) A particle is projected from a point O with a speed of $V \text{ m/s}$ at an angle of θ to the horizontal. Air resistance is to be neglected and $g \text{ m/s}^2$ is the acceleration due to gravity.



- (i) Starting from $\ddot{x} = 0$ show that $x = Vt \cos \theta$ [2]
- (ii) Starting from $\ddot{y} = -g$ show that $y = \frac{-1}{2}gt^2 + Vt \sin \theta$ [2]
- (iii) Prove that the Cartesian equation of path of projectile is given by:

$$y = \frac{-gx^2}{2V^2}(1 + \tan^2 \theta) + x \tan \theta$$

- (iv) You are given that $V^2 = 8g$ and that the particle passes through a point $P(4, 3)$. Hence by using the equation in (iii) find θ , the initial angle of projection and R , the range of the projectile. [2]

END OF PAPER

Q2. a) $\int_0^1 \frac{x}{1+x} dx$

Let $u = 1+x$

$\frac{du}{dx} = 1$

$du = dx$

When $x=1$, $u=2$

When $x=0$, $u=1$

$\int_0^1 \frac{x}{1+x} dx$

$= \int_1^2 \frac{u-1}{u} du$ (1)

$= \int_1^2 (1 - \frac{1}{u}) du$

$= [u - \log u]_1^2$ (1)

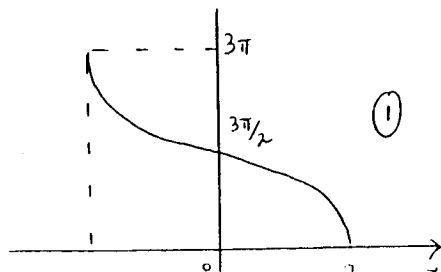
$= 2 - \log 2 - 1 + \log 1$

$= 1 - \log 2.$ (1)

b) i) Domain is $-2 \leq x \leq 2$

Range is $0 \leq y \leq 3\pi$. (1)

ii) $y = 3 \cos^{-1}(\frac{x}{2})$



c) i) If $t = \tan \frac{A}{2}$ then

$\sin A = \frac{2t}{1+t^2}$ (1)

$\cos A = \frac{1-t^2}{1+t^2}$ (1)

ii) Prove $\frac{1 + \cos 2A}{\sin 2A} = \cot A$.

LHS = $\frac{1 + (2\cos^2 A - 1)}{2\sin A \cos A}$

$= \frac{2\cos^2 A}{2\sin A \cos A}$

$= \frac{\cos A}{\sin A} = \cot A = \text{RHS.}$

OR:

LHS = $\frac{1 + \frac{1-t^2}{1+t^2}}{\frac{2t}{1+t^2}}$

$= \frac{(1+t^2) + (1-t^2)}{2t}$ (1)

$= \frac{1}{t} = \frac{1}{\tan A} = \cot A = \text{RHS.}$ (1)

d) i) $y = \log_e(x-1)$ (1)

ii) $y = -\log_e x$ (1)

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PSK

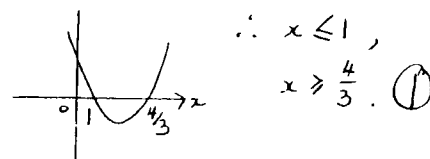
Q1. a) $\frac{1}{x-1} \leq 3$

$x-1 \leq 3(x-1)^2$ (1)

$0 \leq 3(x-1)^2 - (x-1)$

$0 \leq (x-1)[3(x-1)-1]$

$0 \leq (x-1)(3x-4)$ (1)



b) $\int_1^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$

$= [\sin^{-1} \frac{x}{2}]_1^{\sqrt{3}}$ (1)

$= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} \frac{1}{2}$

$= \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}$ (1)

c) $y = \tan^{-1} 2x$

$y' = \frac{1}{1+4x^2} \cdot 2 = \frac{2}{1+4x^2}$

d) $\int_0^{\pi/6} \sin^2 2x dx$

$= \int_0^{\pi/6} \frac{1}{2}(1 - \cos 4x) dx$ (1)

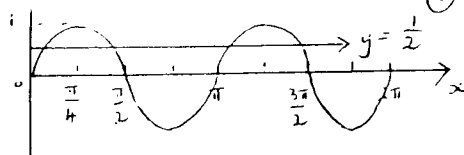
$= \frac{1}{2} [x - \frac{1}{4} \sin 4x]_0^{\pi/6}$

$= \frac{1}{2} [\frac{\pi}{6} - \frac{1}{4} \sin \frac{2\pi}{3}]$ (1)

$= \frac{1}{2} [\frac{\pi}{6} - \frac{1}{4} \cdot \frac{\sqrt{3}}{2}]$

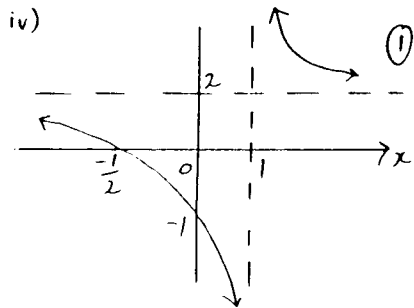
$= \frac{\pi}{12} - \frac{\sqrt{3}}{16} = \frac{4\pi - 3\sqrt{3}}{48}$ (1)

e) i) $y = \sin 2x$



ii) 4 solutions. (1)

- i) Domain is all $x, x \neq 1$. ①
 ii) As $x \rightarrow \infty, y \rightarrow 2$. ①
 iii) Vertical asy. at $x = 1$. ①
 Horiz. asy. at $y = 2$. ①



b) The R th term in the expansion of $(2x^3 - \frac{1}{x})^{12}$ is

$$\begin{aligned}
 & {}^{12}C_{R-1} (2x^3)^{13-R} \left(\frac{-1}{x}\right)^{R-1} \quad ① \\
 & = {}^{12}C_{R-1} \cdot (-1)^{R-1} \cdot (2)^{13-R} \cdot x^{39-3R+1-R} \quad ① \\
 & = {}^{12}C_{R-1} \cdot (-1)^{R-1} \cdot (2)^{13-R} \cdot x^{40-4R} \quad ①
 \end{aligned}$$

This is independent of x if $R = 10$
 ie, $40 - 4R = 0$ is indep. of x . It is

$$\begin{aligned}
 & {}^{12}C_9 \cdot (-1)^9 \cdot 2^3 \\
 & = -1760 \quad ①
 \end{aligned}$$

$$\begin{aligned}
 2(1 - \cos^2 \theta) + \sin^2 2\theta &= 2 \quad ① \\
 2 - 2\cos^2 \theta + \sin^2 2\theta &= 2 \\
 \sin^2 2\theta - 2\cos^2 \theta &= 0 \quad ① \\
 (\sin 2\theta - \sqrt{2} \cos \theta)(\sin 2\theta + \sqrt{2} \cos \theta) &= 0 \\
 \cos^2 \theta (2\sin \theta - \sqrt{2})(2\sin \theta + \sqrt{2}) &= 0 \\
 \cos \theta = 0 \text{ or } \cos \theta = \pm \frac{1}{\sqrt{2}} \quad ① \\
 \theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4} \quad ①
 \end{aligned}$$

has $m_1 = 2$. ①
 $y = -3x + 7$ has $m_2 = -3$.
 Let α be angle between these lines, then
 $\tan \alpha = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$
 $= \left| \frac{2 - (-3)}{1 + 2(-3)} \right| = 1 \quad ①$
 $\therefore \alpha = 45^\circ \quad ①$

b) $v^2 = 16x - 4x^2 + 20$

$$\begin{aligned}
 \text{i) } \frac{1}{2} v^2 &= 8x - 2x^2 + 10 \\
 \frac{d}{dx} \left(\frac{1}{2} v^2 \right) &= 8 - 4x \quad ① \\
 &= -4(x - 2) \\
 &= -2^2(x - 2) \quad ①
 \end{aligned}$$

Since $\ddot{x}$ is of the form
 $\ddot{x} = -n^2 X, X = x - 2$,
 then motion is SHM.

- ii) Centre is $x = 2$. ①
 iii) length of path is 6 units. ①

$$\begin{aligned}
 A &= 4\pi R^2 \\
 \frac{dA}{dR} &= 8\pi R \\
 \frac{dA}{dt} &= \frac{dA}{dR} \cdot \frac{dR}{dt} \\
 &= 8\pi R \cdot \frac{dR}{dt} \quad ①
 \end{aligned}$$

To find $\frac{dR}{dt}$:

$$V = \frac{4}{3} \pi R^3$$

$$\frac{dV}{dR} = 4\pi R^2$$

$$\frac{dV}{dt} = \frac{dV}{dR} \cdot \frac{dR}{dt} \quad ①$$

$$50 = 4\pi R^2 \cdot \frac{dR}{dt}$$

$$\frac{dR}{dt} = \frac{50}{4\pi R^2} = \frac{25}{2\pi R^2} \quad ①$$

Hence

$$\frac{dA}{dt} = 8\pi R \cdot \frac{25}{2\pi R^2} \quad ①$$

$$= \frac{100}{R}$$

When $R = 20$,

$$\frac{dA}{dt} = 5 \text{ mm}^2/\text{sec.} \quad ①$$

Q5. a)

$$\rightarrow \text{iii) } m \text{ of } OQ = \frac{aq^2 - 0}{2aq - 0}$$

$$= \frac{q}{2}$$

$$m \text{ of } OP = \frac{ap^2 - 0}{2ap - 0}$$

$$= \frac{p}{2}$$

Since $\angle QOP = 90^\circ$ then

$$\frac{q}{2} \times \frac{p}{2} = -1$$

$$\text{ie, } pq = -4 \quad (1)$$

$$\rightarrow \text{i) } y = \frac{1}{4a} x^2$$

$$f'(x) = \frac{x}{2a}$$

$$f'(2ap) = \frac{2ap}{2a} = p$$

Hence eq'n of tangent at P is

$$y - ap^2 = p(x - 2ap) \quad (1)$$

$$y - ap^2 = px - 2ap^2$$

$$\text{iv, } y - px + ap^2 = 0 \quad (1)$$

$$\rightarrow \text{ii) Solve } \begin{cases} y - px + ap^2 = 0 \\ y - qx + aq^2 = 0 \end{cases}$$

$$(1) - (2): -px + qx + ap^2 - aq^2 = 0$$

$$x(q - p) = a(q^2 - p^2) \quad (1)$$

$$\text{ie, } x = a(q + p)$$

$$\therefore y = apq \quad (1)$$

$$R \text{ is } \{a(p+q), apq\}$$

$$\text{iv) } x = a(p+q)$$

$$y = apq$$

$$\text{But } pq = -4, \text{ so}$$

$$y = -4a$$

is locus of R. (1)

b) STEP 1 When $n=1$,

$$\text{LHS} = 1^2 = 1$$

$$\text{RHS} = \frac{1}{6} \cdot 1(2)(3) = 1 \quad (1)$$

Hence the proposition is true when $n=1$.

STEP 2 We assume that

$$1^2 + 2^2 + \dots + k^2 = \frac{1}{6} k(k+1)(2k+1)$$

is true, and now show that

$$1^2 + 2^2 + \dots + k^2 + (k+1)^2 = \frac{1}{6} (k+1)(k+2)(2k+3) \quad (1)$$

$$= \frac{1}{6} (k+1)(k+2)(2k+3)$$

$$\text{Now LHS} = \frac{1}{6} k(k+1)(2k+1) + (k+1)^2$$

$$= \frac{1}{6} (k+1) [k(2k+1) + 6(k+1)]$$

$$= \frac{1}{6} (k+1) [2k^2 + 7k + 6]$$

$$= \frac{1}{6} (k+1)(k+2)(2k+3) \quad (1)$$

$$= \text{RHS.}$$

STEP 3. We have shown that the prop. is true when $n=1$. Assuming that the prop. true when $n=k$, we have shown that prop. is true for $n=k+1$ also. Hence the prop. is true when $n=2$, and so true for $n=3$ and so on. Hence it's true for all natural numbers. (1)

Q6. a)

$$\text{i) } {}^5C_3 \times {}^{10}C_8 = 450 \quad (1)$$

ii) With no restrictions, there are

$${}^{15}C_{11} = 1365 \text{ eleven. } (1)$$

The number with only one regular bowler is

$${}^5C_1 \times {}^{10}C_{10} = 5$$

Hence required probability is

$$\frac{5}{1365} = \frac{1}{273} \quad (1)$$

iii) $\otimes$ Probability

$$= \frac{{}^5C_3 \times {}^{10}C_8 + {}^5C_4 \times {}^{10}C_7 + {}^5C_5 \times {}^{10}C_6}{1365} \quad (1)$$

$$= \frac{1260}{1365} = \frac{12}{13} \quad (1)$$

$$\text{b) i) } (1+x)^n \quad (1)$$

$$= {}^nC_0 x^0 + {}^nC_1 x^1 + \dots + {}^nC_n x^n$$

ii) let $x=1$, then

$$\text{LHS} = 2^n$$

$$\text{RHS} = ({}^nC_0) + ({}^nC_1) + ({}^nC_2) + \dots + ({}^nC_n) \quad (1)$$

Next, let $x=-1$, then

$$\text{LHS} = 0$$

$$\text{RHS} = ({}^nC_0) - ({}^nC_1) + ({}^nC_2) - ({}^nC_3) + \dots \quad (1)$$

$$\text{ie, } ({}^nC_1) + ({}^nC_3) + \dots = ({}^nC_0) + ({}^nC_2) + \dots$$

$$\text{ie, } ({}^nC_1) + ({}^nC_3) + \dots = \frac{1}{2} \times 2^n$$

$$= 2^{n-1} \times 2^n \quad (1)$$

$$= 2^{n-1} \text{ as required}$$

(iii) Consider $(1+x)^n (1+x)^n = (1+x)^{2n}$

Now the coeff. of x^{n+1} is

$$({}^nC_0)({}^nC_1) + ({}^nC_1)({}^nC_0) + ({}^nC_2)({}^nC_{n-2}) + \dots$$

$$+ ({}^nC_n)({}^nC_1)$$

$$= ({}^nC_0)({}^nC_1) + ({}^nC_1)({}^nC_0) + \dots + ({}^nC_n)({}^nC_1) \quad (1)$$

Considering the coeff. of x^{n+1}

from $(1+x)^{2n}$ we get

$${}^{2n}C_{n+1} x^{n+1} = \frac{(2n)!}{(n+1)!(2n-(n+1))!} x^{n+1} \quad (1)$$

Hence

$$({}^nC_0)({}^nC_1) + ({}^nC_1)({}^nC_0) + \dots = \frac{(2n)!}{(n+1)!(n-1)!}$$

as required. (1)

$$\frac{3!}{1!} = 60480 \quad \text{①}$$

$$b) i) \ddot{x} = 0$$

$$\dot{x} = C$$

$$\text{But when } t=0, \dot{x} = V \cos \theta$$

$$\text{so } C = V \cos \theta$$

$$\dot{x} = V \cos \theta$$

$$\therefore x = Vt \cos \theta + C_2$$

$$\text{But } x=0 \text{ when } t=0 \text{ so } C_2=0$$

$$\text{ie, } x = Vt \cos \theta \quad \text{②}$$

$$ii) \ddot{y} = -g$$

$$\dot{y} = -gt + C_3$$

$$\text{But when } t=0, \dot{y} = V \sin \theta$$

$$\text{so } C_3 = V \sin \theta \text{ so}$$

$$\dot{y} = -gt + V \sin \theta$$

$$y = -\frac{gt^2}{2} + Vt \sin \theta + C_4$$

$$\text{But } y=0 \text{ when } t=0 \text{ so } C_4=0$$

$$\text{ie, } y = -\frac{gt^2}{2} + Vt \sin \theta \quad \text{②}$$

$$t = \frac{x}{V \cos \theta} \text{ so,}$$

by substitution,

$$y = -\frac{g}{2} \left(\frac{x}{V \cos \theta} \right)^2 + V \left(\frac{x}{V \cos \theta} \right) \sin \theta$$

$$= -\frac{gx^2}{2V^2} \cdot \frac{1}{\cos^2 \theta} + \frac{x \sin \theta}{\cos \theta}$$

$$= -\frac{gx^2}{2V^2} \cdot \sec^2 \theta + x \tan \theta$$

$$= -\frac{gx^2}{2V^2} (1 + \tan^2 \theta) + x \tan \theta$$

as required. ②

iv) From (iii)

$$3 = \frac{-g \cdot 4^2}{2 \cdot 8g} (1 + \tan^2 \theta) + 4 \tan \theta$$

$$3 = -1(1 + \tan^2 \theta) + 4 \tan \theta \quad \text{①}$$

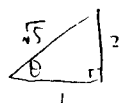
$$\tan^2 \theta - 4 \tan \theta + 4 = 0$$

$$(\tan \theta - 2)(\tan \theta - 2) = 0$$

$$\tan \theta = 2$$

$$\theta = 63^\circ 26'$$

①



To find Range, solve

$$y = 0 \text{ firstly.}$$

$$0 = -\frac{gt^2}{2} + Vt \sin \theta$$

$$\frac{gt^2}{2} = Vt \sin \theta$$

$$t = \frac{2V \sin \theta}{g} \quad \text{①}$$

Sub this into $x = Vt \cos \theta$

and we get

$$R = x = V \left(\frac{2V \sin \theta}{g} \right) \cos \theta$$

$$= \frac{V^2 2 \sin \theta \cos \theta}{g}$$

$$= \frac{V^2 \sin 2\theta}{g} \quad \text{①}$$

$$= \frac{8g \sin 2\theta}{g}$$

$$= 8 \cdot 2 \sin \theta \cos \theta$$

$$= 16 \cdot \frac{2}{\sqrt{5}} \cdot \frac{1}{\sqrt{5}} = \frac{32}{5} \text{ or } 6.4 \text{ m}$$